
Ecological Bias in Multi-Site COVID-19 Studies: Two Failure Modes in 42 Million Patient Records

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Abstract

Multi-site medical studies and federated analysis networks routinely pool site-level summaries to draw conclusions about individual patients. We test this practice using 42 million individual COVID-19 patient records and identify two distinct failure modes of ecological bias.

Failure mode 1: the signal disappears. Using 38 million CDC records partitioned into 9 demographic pseudo-sites, site-level regression cannot detect the relationship between age and mortality ($\beta = +0.28$, $p = 0.12$), while individual-level analysis finds that patients over 80 have 9.9 times the odds of dying ($p < 10^{-300}$).

Failure mode 2: the signal distorts. Using 4 million confirmed COVID-19 cases across Mexico’s 32 states—real geographic sites—site-level regression detects a significant relationship ($\beta = +1.31$, $p < 0.0001$, $R^2 = 0.50$), but the ecological effect size bears no resemblance to the individual-level truth (age 70+ OR = 11.3).

These failures have clinical consequences. Applied to the 4CE consortium’s 22,000-patient neurological COVID-19 dataset (21 hospitals, 6 countries), meta-analytic methods diverge by 350-fold on the pooled effect ($I^2 = 99.8\%$), the proportion of elderly patients explains most between-site variation ($\beta = +14.2$, $p < 10^{-15}$), and a multivariate consistency test detects coordinated cross-site heterogeneity in comorbidity correlations ($z = 25$, $p < 0.0001$) that standard tests miss entirely. Per-site interaction analyses on all three datasets show that the age–comorbidity relationship varies across sites (CDC: 2-fold; Mexico: 2-fold; 4CE: age $\times$ mortality $p < 0.04$ at all timepoints), confirming that pooling loses information about effect modification. When site-level averages fail in two different ways on two different datasets, federated analyses that pool across hospital networks should treat their summaries as hypotheses for individual-level testing.

1 Introduction

Suppose ten hospitals each report the fraction of their COVID-19 patients who died. A meta-analysis averages those fractions and reports the result as “the” mortality rate. The same logic underlies federated analysis networks such as ENACT, TriNetX, and PCORnet, where hospitals share summary statistics rather than individual records. This approach works when the hospitals are measuring the same thing in the same kind of patient.

They rarely are.

Hospitals differ in who walks through the door. One serves a young urban population; another, a rural elderly community. One codes aggressively for neurological complications; another barely records them. The average across these hospitals is a real number, but it may not describe any actual hospital or any actual patient. Worse, it can point in the wrong direction entirely.

This problem—drawing conclusions about individuals from group-level data—is the *ecological fallacy* Robinson (1950). Known since 1950 and taught in every epidemiology course, it is routinely acknowledged and

routinely ignored. Multi-site consortium studies and federated analysis networks typically have access only to site-level summaries and treat them as proxies for patient-level truth.

We test how badly this matters using 42 million individual COVID-19 patient records from two countries, plus a third dataset of 22,000 patients from 21 hospitals across 6 countries. The gap between site-level and individual-level analysis is not subtle. We identify two distinct failure modes:

1. **The signal disappears.** Site-level analysis of 38 million CDC records cannot detect the relationship between age and mortality ($p = 0.12$). Individual-level analysis finds it with overwhelming significance ($OR = 9.9, p < 10^{-300}$).
2. **The signal distorts.** Site-level analysis of 4 million Mexican COVID-19 records finds a significant relationship ($p < 0.0001$), but the ecological effect size ($\beta = +1.31$) bears no resemblance to the individual-level truth ($OR = 11.3$). The regression reports a linear slope; the patient-level reality is a multiplicative odds ratio an order of magnitude larger.

We then ask: if federated networks are constrained to site-level summaries, can anything be done? We show that structural consistency tests—methods that examine the *structure* of relationships across sites, rather than averaging over them—can diagnose when site-level pooling is unreliable. Applied to the 4CE consortium’s neurological COVID-19 dataset, these methods detect coordinated heterogeneity invisible to standard meta-analysis. When site-level averages fail in two different ways on two different datasets, federated analyses should treat their pooled summaries as hypotheses for individual-level testing.

2 Ecological Bias: Two Failure Modes

2.1 Failure mode 1: the signal disappears (CDC, United States)

We start with the simplest possible question: does age predict COVID-19 mortality? The answer is obviously yes. The question is whether site-level analysis can find it.

The CDC Case Surveillance dataset contains 38.2 million confirmed COVID-19 cases with age group, sex, race/ethnicity, hospitalization status, ICU admission, and death. We used race/ethnicity categories as pseudo-sites (9 groups with $n \geq 1,000$), mimicking what a consortium study would have if each racial/ethnic group were a separate hospital reporting summary statistics.

Site-level analysis aggregated each group’s mortality rate and proportion of elderly patients, then regressed one on the other across the 9 groups. Result: no significant relationship ($\beta = +0.28, p = 0.12, R^2 = 0.31$). Site-level analysis cannot detect that age predicts mortality.

Individual-level analysis on the same 38.2 million patients found that being over 80 multiplies the odds of death by 9.9 ($p < 10^{-300}$). After adjusting for sex and comorbidities, patients gain 2.4 times the odds of dying for each additional decade of age ($p < 10^{-300}$). Males have 1.7 times the odds; patients with underlying conditions have 1.5 times the odds.

The effect is overwhelming at the individual level and invisible at the site level. This is the ecological fallacy at its most extreme: not merely distorting the effect, but eliminating it entirely.

2.2 Failure mode 2: the signal distorts (Mexico)

The CDC demonstration uses pseudo-sites (racial/ethnic groups), which one might dismiss as an artifact of the grouping choice. Mexico’s COVID-19 surveillance data provides a stronger test: 3,993,464 confirmed cases across 32 real geographic states, each corresponding to the state of the treating medical unit.

Site-level analysis across the 32 states finds a significant relationship between the proportion of elderly patients and the state mortality rate ($\beta = +1.31, p < 0.0001, R^2 = 0.50$). The ecological regression “works”—it detects a real signal, unlike the CDC analysis.

Table 1: Two failure modes of ecological bias. The CDC analysis (pseudo-sites) *misses* the age–mortality relationship entirely. The Mexico analysis (real geographic sites) *detects* a relationship but in the wrong functional form and at the wrong scale. Individual-level analysis on both datasets finds overwhelming effects.

Dataset	Analysis level	Effect size	p -value
<i>CDC (38.2M cases, 9 pseudo-sites):</i>			
	Site-level regression	$\beta = +0.28$	0.12
	Individual: age per decade	OR = 2.40	$< 10^{-300}$
	Individual: age 80+	OR = 9.88	$< 10^{-300}$
<i>Mexico (4.0M cases, 32 states):</i>			
	Site-level regression	$\beta = +1.31$	< 0.0001
	Individual: age 70+	OR = 11.26	$< 10^{-300}$
	Individual: adjusted	OR = 7.97	$< 10^{-300}$

Individual-level analysis on the same 4 million patients reveals the true effect: patients aged 70 or older have 11.3 times the odds of dying ($p < 10^{-300}$). After adjusting for sex and 9 individual comorbidities (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, and smoking), the adjusted odds ratio remains 8.0 ($p < 10^{-300}$). Males have 1.7 times the odds; patients with any comorbidity have 3.8 times the odds.

The ecological regression reports a linear slope of +1.31; the patient-level reality is a multiplicative odds ratio an order of magnitude larger. The site-level analysis detects the direction of the effect, but the magnitude it reports—a regression coefficient in the wrong functional form—is incommensurable with the individual-level truth.

These two failure modes are complementary. The CDC data shows that ecological analysis can be blind to effects that are overwhelming at the patient level. The Mexico data shows that even when ecological analysis detects a signal, the reported effect size can be incommensurable with individual-level truth—a linear slope versus a multiplicative odds ratio 11 times the baseline. Any federated analysis relying on site-level summaries faces both risks simultaneously.

2.3 Why does this happen?

The ecological fallacy arises from confounding at the group level. To see the mechanism, consider the CDC data.

The group with the highest proportion of 80+ patients (“Unknown” race, 13.5%) has lower mortality (3.3%) than the group with the second-highest elderly proportion (Hispanic, 9.4%, mortality 7.6%), because the Unknown group has different sex composition and comorbidity rates. Groups also differ in sex ratios (15% to 65% male) and comorbidity prevalence (0.5% to 14%). These differences scramble the age–mortality relationship at the group level.

At the individual level, confounders are controlled directly: the logistic regression asks “among patients with the same sex and comorbidity status, does age predict death?” The answer is unambiguous.

Mexico illustrates the second mechanism. The 32 states have less extreme compositional variation than the CDC pseudo-sites (the states are real geographic units, not demographic slices), so the ecological regression does detect a directional signal. The distortion arises from aggregation bias: the regression fits a linear slope to state-level proportions, while the true patient-level relationship is a nonlinear odds ratio. Simpson’s paradox does not apply in the classic sense—the direction is preserved—but the magnitude and functional form are lost entirely.

Table 2: Meta-analysis of CNS neurological involvement on COVID-19 severity (PMI) at 30 days. All three estimates are unreliable for different reasons. The 350-fold divergence is a diagnostic signal that no single pooled number is appropriate for this data.

Method	Pooled PMI	95% CI	p	I^2
Fixed-effects (IV)	1.97	[1.87, 2.09]	$< 10^{-6}$	99.8%
Random-effects (DL)	702	[201, 2460]	$< 10^{-6}$	—
Influence function	972	[1.6, 587K]	0.035	—

These are not subtle statistical points. They mean that any federated analysis reporting pooled site-level averages could be reporting the wrong magnitude, the wrong functional form, or no effect at all—depending on the specific confounding structure of the network’s sites.

3 Clinical Consequences: The 4CE Neurological Dataset

The ecological bias demonstrated in Sections 2.1–2.2 has direct clinical consequences. The 4CE consortium (Consortium for Clinical Characterization of COVID-19 by EHR) analyzed neurological outcomes in over 22,000 hospitalized patients across 21 hospitals in 6 countries Chou et al. (2021), using only aggregate site-level data. We re-analyze this dataset to show that its pooled estimates are unreliable in precisely the ways the ecological fallacy predicts.

3.1 CNS involvement and the 350-fold method divergence

Among well-measured sites ($SE < 0.5$ on the log scale; $k = 11$ – 16 per timepoint), the fixed-effects proportional morbidity index (PMI) for CNS involvement is 1.97 (95% CI: [1.87, 2.09]) at 30 days. This estimate must be interpreted cautiously: $I^2 = 99.8\%$ (Cochran’s $Q = 6318$, $p < 10^{-15}$) means essentially all variation is between-site, violating the common-effect assumption that fixed-effects requires.

The 350-fold divergence between pooling methods (Table 2) is a diagnostic failure. The random-effects PMI of 702 is a numerical artifact: several small sites produce PMIs of 10^8 – 10^{18} from complete separation, and the random-effects framework up-weights these extreme estimates. Among the 11 sites with $SE < 0.4$, the picture is more consistent (PMIs between 1.4 and 2.8), but even this restricted set shows $I^2 > 95\%$. The method divergence signals that this dataset requires decomposition of heterogeneity rather than a pooled estimate.

Peripheral nervous system involvement provides a negative control. PNS diagnoses show a PMI of 1.04 at 30 days (95% CI: [0.95, 1.14], $p = 0.37$), indistinguishable from the null. If the CNS effect were driven by sicker patients receiving more diagnostic workups (which would also detect peripheral neuropathies), PNS diagnoses should show a similar signal. They do not. The specificity to CNS involvement is consistent with SARS-CoV-2 neuroinvasion through the blood-brain barrier Song et al. (2021) and central neuroinflammation Iadecola et al. (2020), mechanisms that do not produce peripheral neuropathy.

3.2 Age drives between-site variation

Meta-regression of the CNS severity effect (30-day PMI, log scale) on four site-level demographics identifies the proportion of elderly patients as the strongest moderator ($\beta = +14.2$, $SE = 0.83$, $p < 10^{-15}$). Each 10-percentage-point increase in the share of patients aged 80+ corresponds to a $\exp(1.42) \approx 4.1$ -fold increase in the neurological severity PMI. Sites serving older populations see dramatically larger neurological effects.

This pattern is consistent with age-dependent blood-brain barrier vulnerability Montagne et al. (2015) and chronic low-grade inflammation Franceschi et al. (2018), but site-level data cannot distinguish this biological explanation from compositional confounding: sites with more elderly patients also differ in comorbidity burden, coding practices, and clinical management.

Table 3: Meta-regression moderators of the CNS severity association (30-day PMI). Elderly proportion is the strongest moderator. Baseline severity acts in the opposite direction (ceiling effect). The large β magnitudes are likely inflated by Type M error Gelman & Carlin (2014) from ~ 20 sites.

Moderator	β	SE	p	Range across sites
Proportion age ≥ 80	+14.2	0.83	$< 10^{-15}$	0.01–0.43
Proportion female	+1.13	0.14	$< 10^{-15}$	0.06–0.53
Baseline severity	−2.91	0.23	$< 10^{-15}$	0.06–0.73
Number of patients	−0.0003	0.00005	$< 10^{-9}$	32–22,789

Table 4: Cross-site consistency of comorbidity correlation patterns. The multivariate consistency statistic detects collective inconsistency ($z = 25$) that pairwise Cochran’s Q tests miss entirely (0/1305 significant).

Test	Statistic	p	Significant pairs
Consistency (all patients)	$z = 25.0$	< 0.0001	N/A (collective)
Consistency (CNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Consistency (PNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Cochran’s Q (per pair)	$\max Q = 13.4$	all > 0.05	0/1305

A site-level interaction between elderly proportion and mortality rate is significant at all three follow-up periods ($p < 0.04$ uncorrected), increasing from 30 to 90 days (β : $+32.5 \rightarrow +38.3$), though the temporal trend itself has only three timepoints and does not reach significance ($p = 0.16$). Under Benjamini–Hochberg FDR control for 18 tests, only the 60- and 90-day values survive; the 30-day value ($p = 0.037$) is suggestive but does not survive correction. The coefficient magnitudes ($\beta = +32$ to $+38$) are characteristic of Type M inflation Gelman & Carlin (2014): estimated from ~ 20 aggregated sites with high variance, significant estimates are systematically exaggerated. A hierarchical meta-regression with site random effects would shrink these estimates toward the group mean Gelman et al. (2012).

3.3 Comorbidity correlations differ systematically across sites

Elixhauser comorbidity correlation matrices Elixhauser et al. (1998) from 20 4CE sites show collective inconsistency when tested with a multivariate consistency statistic: $z = 25.0$ ($p < 0.0001$), with comparable signals in CNS-only ($z = 14.4$) and PNS-only ($z = 14.4$) subgroups (Table 4).

No single comorbidity pair varies enough across sites to reach significance on its own. The consistency test detects their collective pattern: the aggregate structure of small per-pair differences is far more extreme than chance. This matters because standard meta-analysis assumes the covariate relationships being adjusted for are constant across sites. The most heterogeneous pairs—HIV-Lymphoma ($Q = 13.4$, $r = 0.09 \pm 0.25$) and Alcohol-Drugs ($Q = 12.9$, $r = 0.70 \pm 0.24$)—illustrate how coding and documentation practices drive cross-site inconsistency that the collective test detects.

4 Structural Diagnostics Across All Three Datasets

When individual data is unavailable, can federated networks diagnose whether their site-level summaries are trustworthy? Two structural diagnostics—the consistency statistic and per-site interaction analysis—provide partial answers.

4.1 The consistency test: power requires heterogeneous coverage

The consistency statistic measures whether per-site correlation matrices disagree more than expected by chance. Its power depends on a structural property of the data: whether different sites report different subsets of variable pairs (heterogeneous coverage) or all report the same pairs (homogeneous coverage).

On the CDC data (9 groups, all reporting all 15 variable pairs), the permutation null is degenerate: $z = 0$, null standard deviation $< 10^{-16}$. On Mexico’s 32 states (all reporting all 15 pairs), the same degeneracy appears: $z = 0.47$, effectively null. Appendix A proves this is a mathematical necessity: within-dimension permutation preserves the test statistic when coverage is homogeneous.

On the 4CE data, where coverage is heterogeneous (10 sites report 36–296 of 1,305 possible comorbidity pairs; 10 report all 1,305), the same test produces $z = 25$ ($p < 0.0001$). The test’s power depends on heterogeneous coverage—a property of the data structure, not the signal. This is a practical guide for federated networks: the consistency test is most useful precisely when sites have different measurement capabilities, which is the norm in multi-national consortia.

4.2 Per-site interactions: effect modification varies everywhere

Per-site logistic regressions reveal genuine variation in the age $\times$ comorbidity interaction across all three datasets.

CDC (9 pseudo-sites). All groups show that comorbidity matters less in the elderly (all interaction OR < 1), but the magnitude varies 3-fold: from OR = 0.23 (Missing) to OR = 0.69 (Hispanic/Latino), all significant ($p < 10^{-4}$). Dropping the two groups with ambiguous demographic composition (Unknown and Missing) leaves 7 groups spanning 0.27–0.69, a 2.6-fold range. The pooled individual-level regression confirms the interaction is real (OR = 0.92 per decade, $p < 10^{-132}$).

Mexico (32 states). The same pattern holds with real geographic sites: pooled interaction OR = 0.32 ($p < 10^{-300}$), with per-state values ranging from 0.19 (Quintana Roo) to 0.38 (Mexico City), a 2-fold range. All 32 states show sub-additive interactions—comorbidity adds less mortality risk in the elderly—but the magnitude of this attenuation varies by state.

4CE (21 hospitals). The age $\times$ mortality interaction on the neurological severity effect is significant at all three timepoints ($p < 0.04$ uncorrected), with coefficients that increase from $\beta = +32.5$ at 30 days to $\beta = +38.3$ at 90 days.

The consistency of per-site interaction variation across three independent datasets—with different countries, different site definitions, and different outcome measures—is direct evidence that pooling the age–comorbidity or age–mortality relationship across sites loses information about effect modification.

5 Discussion

5.1 Two failure modes, one underlying problem

The CDC and Mexico analyses demonstrate that ecological bias in federated studies is not a single phenomenon with a single severity. It has at least two distinct failure modes, determined by the confounding structure of the network’s sites.

When confounders vary strongly across sites relative to the exposure–outcome relationship (the CDC case, where race/ethnicity groups differ dramatically in sex and comorbidity composition), the ecological regression has no power to detect an effect that is overwhelming at the individual level. When confounders are more balanced across sites (the Mexico case, where states are real geographic units with more homogeneous demographic variation), the ecological regression detects the direction of the effect but reports it in the wrong functional form and at the wrong scale. The first mode is more dangerous because it produces a false negative; the second is more insidious because it produces apparent confirmation of a relationship whose magnitude is meaningless.

Table 5: Summary of methods applied across all three datasets. The consistency statistic has power only on heterogeneous-coverage data (4CE). Per-site interaction analysis detects effect modification on all three datasets. Individual-level analysis (where available) gives the true answer.

Method	Dataset	Statistic	p
Ecological regression	CDC (9 groups)	$\beta = +0.28$	0.12
	Mexico (32 states)	$\beta = +1.31$	< 0.0001
	4CE (21 hospitals)	PMI = 2.0	$< 10^{-6}$
Consistency statistic	CDC	degenerate	—
	Mexico	degenerate	—
	4CE	$z = 25$	< 0.0001
Per-site interaction	CDC	OR: 0.23–0.69	all $< 10^{-4}$
	Mexico	OR: 0.19–0.38	all $< 10^{-4}$
	4CE	β : +32 to +38	< 0.04
Individual logistic	CDC	OR = 9.88	$< 10^{-300}$
	Mexico	OR = 11.26	$< 10^{-300}$

Both failure modes have the same root cause: aggregation destroys the within-site covariance structure that individual-level analysis exploits. The practical implication for federated networks such as ENACT, TriNetX, and PCORnet is that site-level summaries should never be treated as sufficient statistics for the patient-level relationships they purport to represent.

5.2 Implications for federated causal discovery

Federated analysis networks face a structural tension. Privacy constraints and data governance restrict them to site-level summaries, but ecological bias means these summaries can miss, distort, or misattribute patient-level effects. The consistency test and per-site interaction analysis offer partial diagnostics: the consistency test detects coordinated heterogeneity when coverage is heterogeneous, and per-site interaction analysis detects effect modification regardless of coverage structure.

These tools are diagnostic, not corrective. They can warn that site-level pooling is unreliable, but they cannot recover the individual-level answer. Three approaches can:

Federated regression. Sites run a standardized analysis script locally and share only coefficients (point estimates, standard errors, and sample sizes). This preserves patient privacy while enabling individual-level estimation via meta-analysis of patient-level effects rather than meta-analysis of ecological summaries.

Large individual-level cohorts. N3C (15 million patients, 75+ hospital sites, full EHR) and ISARIC (705,000 hospitalized patients, 1,500+ sites, 60 countries) contain the individual records needed to directly test whether neurological involvement predicts severity after patient-level adjustment.

Hierarchical models with site random effects. When individual-level data is available, a hierarchical model with site as a random intercept directly addresses both the ecological fallacy (by estimating patient-level effects) and the Type M inflation problem (by shrinking extreme site-level estimates toward the group mean Gelman et al. (2012)). The shrinkage factor is approximately $\hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_i^2)$, where $\hat{\tau}^2$ is the between-site variance and $\hat{\sigma}_i^2$ is the within-site sampling variance.

5.3 Practical guidelines for multi-site studies

The ecological fallacy is not specific to COVID-19. Any multi-site study that pools site-level summaries—genome-wide association studies, multi-center drug trials, educational interventions, crime statistics—faces the same risk when the units of analysis (sites) differ systematically from the units of inference (people).

Our results across three datasets suggest five guidelines:

- 1. Report heterogeneity honestly.** $I^2 = 99.8\%$ means the pooled average is almost meaningless. When heterogeneity exceeds 90%, present the full distribution of site-level effects alongside the average, with explicit acknowledgment that the average may not apply to any individual site.
- 2. Test the structure, not just the average.** Consistency tests detect when relationships between variables differ across sites in coordinated ways—information that Cochran’s Q (which tests one variable at a time) and geometric distance methods (which detect expected demographic variation) both miss.
- 3. Use hierarchical models.** A single hierarchical model with site as a random effect addresses the multiplicity problem at the model level (through partial pooling) and the Type M problem directly (through shrinkage of extreme estimates).
- 4. Disclose the full analysis menu.** State how many tests were conducted and report all results, including nulls. Undisclosed analytic flexibility is what destroys credibility.
- 5. Seek individual data.** The ecological fallacy is a problem of aggregation; the only definitive solution is disaggregation. Modern data-sharing frameworks (federated analysis, secure enclaves, synthetic data) make individual-level analysis feasible even when raw data cannot leave the hospital.

6 Methods

6.1 Data sources

CDC Case Surveillance. The CDC COVID-19 Case Surveillance Public Use Dataset contains 38,173,454 confirmed and probable COVID-19 cases reported to the CDC through 2022. Each record includes onset date, age group (9 categories: 0–9 through 80+), sex, race/ethnicity (9 categories), hospitalization, ICU admission, death, and underlying medical condition (binary flag). No geographic identifiers are present in the public version. We used race/ethnicity categories as pseudo-sites, restricted to groups with $n \geq 1,000$ (9 groups, n ranging from 13,548 to 6,165,036).

Mexico Datos Abiertos. Mexico’s Secretaría de Salud publishes individual-level COVID-19 surveillance data covering all confirmed and suspected cases nationwide. We used the January 3, 2022 historical snapshot (12,425,179 total records, 3,993,464 confirmed COVID-19 cases, 299,581 deaths). Each record includes age, sex, state of the treating medical unit (ENTIDAD_UM, 32 states), death date, and 9 individual comorbidity flags (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, smoking). Cases were classified as confirmed when CLASIFICACION_FINAL $\in \{1, 2, 3\}$. We used state of medical unit as the site variable, providing 32 real geographic sites with n ranging from 8,759 (Colima) to 637,408 (Mexico City).

4CE Consortium. Aggregate site-level data from the 4CE consortium’s Phase 2.1 Neurological Analysis Chou et al. (2021): 21 healthcare sites across 6 countries (USA, France, Germany, UK, Singapore, Italy), over 22,000 hospitalized COVID-19 patients. Data included site-level demographics, Elixhauser comorbidity scores Elixhauser et al. (1998), and proportional morbidity index (PMI) for CNS and PNS neurological involvement at 30, 60, and 90 days.

6.2 Statistical analyses

Ecological regression. For each dataset, we computed per-site proportions of elderly patients and mortality rates, then regressed site mortality on site elderly proportion using ordinary least squares. For CDC data, elderly was defined as age 80+; for Mexico, age 70+.

Individual-level logistic regression. We fit logistic regression models on the complete individual-level data: (1) $\text{death} \sim \text{age}$; (2) $\text{death} \sim \text{age} + \text{sex} + \text{comorbidity}$; (3) $\text{death} \sim \text{age} \times \text{comorbidity} + \text{sex}$. For the CDC data, age was coded as the midpoint of the age group (5, 15, ..., 85); for Mexico, age was continuous, dichotomized at 70. Odds ratios are reported per decade (CDC) or as binary contrasts (Mexico). We used `statsmodels` v0.14 for estimation.

Per-site interaction analysis. For each site in each dataset, we fit a logistic regression with an age $\times$ comorbidity interaction term. For the 4CE data (no individual records), we fit weighted meta-regressions with interaction terms on site-level demographics.

Multivariate consistency test. We computed per-site Pearson correlation matrices (6 binary variables for CDC and Mexico; 30 Elixhauser comorbidity categories for 4CE) and measured global consistency as the mean sum of squared pairwise differences across all site pairs, normalized by shared coverage. Significance was assessed by 500 within-dimension permutations. The test’s power depends on heterogeneous coverage (Appendix A).

4CE meta-analysis. Per-site log-PMI estimates were pooled using fixed-effects (inverse-variance), DerSimonian–Laird random-effects, and influence-function methods. Sites with $\text{SE} \geq 0.5$ were excluded from primary analyses. Meta-regression of log-PMI on four site-level demographics used weighted least squares. E-values VanderWeele & Ding (2017) assess sensitivity to unmeasured confounding.

6.3 Reproducibility

All code and analysis scripts are available at <https://github.com/TODO/ecological-fallacy-covid> and archived at <https://doi.org/10.5281/zenodo.TODO>. The CDC data is publicly downloadable from data.cdc.gov (dataset vbim-akqf). The Mexico data is publicly downloadable from datos.gob.mx (Dirección General de Epidemiología).

7 Limitations

Pseudo-sites are not hospitals. Race/ethnicity groups in the CDC data are a proxy for the multi-site structure of consortium studies. They differ from hospitals in important ways: they are not geographically localized, patients within a group may attend different hospitals, and the grouping confounds race with socioeconomic factors. The ecological fallacy demonstration is valid—group-level analysis misses patient-level effects—but the specific confounders differ from those in a hospital-based consortium. The Mexico analysis (32 real geographic sites) partially addresses this limitation, and the two datasets together bracket the range of ecological bias severity.

Limited variables. The CDC public dataset has 12 variables; the Mexico data has 18. Hospital-based datasets such as N3C have hundreds. Our analysis captures the direction of the problem but underestimates its magnitude: with more variables, the scope for ecological confounding grows.

4CE data is aggregate only. All 4CE analyses use site-level summaries. The ecological fallacy applies directly: site-level associations need not hold for individual patients, and the age–mortality interaction may reflect compositional confounding rather than a biological mechanism. The age $\times$ mortality interaction coefficients ($\beta = +32$ to $+38$) are likely inflated by Type M error from ~ 20 sites Gelman & Carlin (2014).

Consistency tests are diagnostic, not corrective. The multivariate consistency test detects *that* something is wrong with site-level pooling; it does not tell you *what* the individual-level answer is. Its power requires heterogeneous coverage—a property present in the 4CE data but absent in the CDC and Mexico data, where the test is degenerate (Appendix A).

Multiple testing. We applied multiple methods across three datasets and (for 4CE) three timepoints. We report all results, including null findings (PNS PMI = 1.04; consistency test degenerate on CDC and Mexico). FDR correction is applied where applicable; some individually significant findings do not survive correction.

8 Conclusion

Ecological bias in federated COVID-19 analysis operates through two distinct failure modes. Site-level analysis of 38 million CDC records cannot detect an age–mortality relationship that is overwhelming at the individual level (OR = 9.9). Site-level analysis of 4 million Mexican records detects the direction of the relationship but reports it at the wrong scale and in the wrong functional form (a linear slope of +1.31 instead of a multiplicative odds ratio of 11.3). The 4CE consortium’s neurological findings—where meta-analytic methods diverge by 350-fold and elderly proportion explains most between-site variation—illustrate the clinical stakes: pooled estimates from this dataset should be treated as hypotheses, not as established effects.

Structural diagnostics offer partial help. The consistency test detects coordinated heterogeneity on the 4CE data ($z = 25$) but is degenerate on homogeneous-coverage datasets (CDC, Mexico). Per-site interaction analysis detects effect modification across all three datasets, confirming that pooling loses information about how age and comorbidity interact.

The practical recommendation: when heterogeneity is extreme, do not trust the average. When individual data is unavailable, test the structure. When neither is possible, report site-level findings explicitly as ecological summaries that may not hold for individual patients—and design the next study to collect individual-level data.

A Coverage-Degeneracy of the Consistency Test

Proposition 1. *When all sites report all variable pairs (homogeneous coverage), within-dimension permutation of the consistency statistic is degenerate: the null distribution has zero variance.*

Proof. Let r_{ij} denote the correlation for variable pair j at site i . The consistency statistic is $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{|S_{ii'}|} \sum_{j \in S_{ii'}} (r_{ij} - r_{i'j})^2$, where $S_{ii'}$ is the set of pairs reported by both sites i and i' . Under homogeneous coverage, $S_{ii'} = \{1, \dots, p\}$ for all site pairs, so $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{p} \sum_{j=1}^p (r_{ij} - r_{i'j})^2$. The within-dimension permutation shuffles $\{r_{1j}, \dots, r_{kj}\}$ independently for each j . For fixed j , any permutation σ preserves $\sum_{i < i'} (r_{\sigma(i),j} - r_{\sigma(i'),j})^2 = k \sum_i r_{ij}^2 - (\sum_i r_{ij})^2$, which depends only on the multiset $\{r_{ij}\}_{i=1}^k$. Since every pair $S_{ii'}$ includes the same set of indices j , the total T is a sum of terms each invariant under the permutation, so T is constant across all permutations. $\square$

When coverage is heterogeneous, permutation changes which site-pair sums include which terms, breaking the invariance. The 4CE data has heterogeneous coverage (10 sites report 36–296 of 1,305 pairs; 10 report all 1,305), producing $z = 25$. Both the CDC pseudo-sites (9 groups, all 15 pairs) and Mexico’s 32 states (all 15 pairs) have homogeneous coverage, producing degenerate permutation nulls ($z \approx 0$, null standard deviation $< 10^{-16}$). The degeneracy on Mexico’s data—which uses real geographic sites, not constructed pseudo-sites—confirms that the limitation is structural (homogeneous coverage), not an artifact of the CDC’s pseudo-site design.

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